

STATS 200A Lecture 1

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Sample space Ω : the possible outcomes. An event is the set of outcomes where a statement holds. For coin tossing, $H_n = \{\text{toss } n \text{ is heads}\}$.

$\Omega = \{0, 1\}^{\mathbb{N}} = \{(\omega_1, \omega_2, \dots, \omega_n, \dots) : \omega_i \in \{0, 1\}\}$

$\omega = (0, 1, 1, 0, 1, 1, 0, \dots), \omega \in H_3$

$\omega' = (1, 0, 0, 1, \dots)$

$H_n = \{\text{toss } n \text{ is a heads}\}$

Not A : A^c

A or B : $A \cup B$

A and B : $A \cap B$

Interesting events are built using *countably many* unions and intersections. The last row is the strong-law event; work through its formula in the homework.

At least one head: $\bigcup_{n \geq 1} H_n$

Heads infinitely often: $\forall N \in \mathbb{N}, \exists n \geq N, \omega \in H_n$

$$\bigcap_{N \geq 1} \bigcup_{n \geq N} H_n$$

Frequency of heads $\rightarrow \frac{1}{2}$

$$\bigcap_{k \geq 1} \bigcup_{N \geq 1} \bigcap_{n \geq N} \left\{ \left| \frac{1}{n} \sum_{i=1}^n \mathbf{1}_{H_i} - \frac{1}{2} \right| < \frac{1}{k} \right\}$$

Definition 1.1

σ -algebra $\mathcal{F} \subseteq 2^\Omega$ is a σ -algebra if

(i) $\Omega \in \mathcal{F}$;

(ii) $A \in \mathcal{F} \Rightarrow A^c \in \mathcal{F}$;

(iii) $A_1, A_2, \dots \in \mathcal{F} \Rightarrow \bigcup_n A_n \in \mathcal{F}, \left(\bigcup_{n \geq 1} A_n^c \right)^c$

$(\Omega, \mathcal{F})$: a *measurable space*. Elements of $\mathcal{F}$: *events*.

Follows: $\emptyset \in \mathcal{F}$; countable intersections (De Morgan); finite unions; differences.
Not required, on purpose: closure under *uncountable* unions.

Example 1.2

Roll a die, $\Omega = \{1, \dots, 6\}$.

$\mathcal{F}_0 = \{\emptyset, \Omega\}$

$\mathcal{F}_1 = \{\emptyset, \{1, 3, 5\}, \{2, 4, 6\}, \Omega\}$

$\mathcal{F}_2$ = all unions of the blocks $\{1, 2\}, \{3, 4\}, \{5, 6\}$

2^Ω : all 64 subsets

Definition 1.3 (atom)

For finite $\mathcal{F}$, the *atom* containing $\omega : \langle \omega \rangle = \bigcap \{A \in \mathcal{F} : \omega \in A\}$, the smallest set $\mathcal{F}$ can see around ω .

Theorem 1.4

If $\mathcal{F}$ is finite, its distinct atoms $A_1, \dots, A_k$ partition Ω , and $\mathcal{F} = \{\bigcup_{i \in I} A_i : I \subseteq \{1, \dots, k\}\}$ SO $|\mathcal{F}| = 2^k$. Conversely every finite partition gives a σ -algebra this way.

Proof. two atoms are equal or disjoint; every $A \in \mathcal{F}$ is $\bigcup_{\omega \in A} \langle \omega \rangle$ ■

Remark 1.5 (A σ -algebra is a resolution)

A partition is a level of detail at which you can look at the outcome. The events in $\mathcal{F}$ are exactly the questions you can answer at that resolution. $\mathcal{F}_1$: you were told the parity. $\mathcal{F}_2$: you were told $\lceil \omega/2 \rceil$. $\{\emptyset, \Omega\}$: you know nothing. 2^Ω : you know everything

Functions you can compute at resolution $\mathcal{F}$ are the ones constant on atoms. If all you know is the block, you can only evaluate quantities that depend on ω through its block, (Wednesday: this is what ‘ $\mathcal{F}$ -measurable’ means for finite $\mathcal{F}$)

Refining the partitions = enlarging the σ -algebra = gaining information. $\mathcal{F}_0 \subseteq \mathcal{F}_1, \mathcal{F}_0 \subseteq \mathcal{F}_2$, but $\mathcal{F}_1 \not\subseteq \mathcal{F}_2$: parity can’t be recovered from $\lceil \omega/2 \rceil$

Remark 1.6

Intersections of σ -algebras are σ -algebras. For $\mathcal{C} \subseteq 2^\Omega$,

$$\sigma(\mathcal{C}) = \bigcap \{\mathcal{G} : \mathcal{G} \text{ a } \sigma\text{-algebra, } \mathcal{C} \subseteq \mathcal{G}\}.$$

This is the smallest σ -algebra containing the basic events in $\mathcal{C}$, $\mathcal{C} = \{A\}$.

$$\sigma(\{A\}) = \{\emptyset, A, A^c, \Omega\}$$

Definition 1.7 (Borel σ -algebra)

$\mathcal{B}(\mathbb{R}) = \sigma(\text{open sets of } \mathbb{R})$. Its members are the *Borel sets*

Remark 1.8 (Useful generating families)

Open intervals; intervals $(a, b]$ with rational $a < b$; half-lines $(-\infty, x], x \in \mathbb{R}$; or just $(-\infty, q], q \in \mathbb{Q}$

All open and closed sets are Borel, as are singletons:

$$\{x\} = \bigcap_{n \geq 1} (x - 1/n, x].$$

Yet $\mathcal{B}(\mathbb{R}) \neq 2^\mathbb{R}$. Having every singleton does not mean having *every union* of singletons; closure is only required for countable unions.

Definition 1.9 (Measures)

A *measure* on $(\Omega, \mathcal{F})$ is $\mu : \mathcal{F} \rightarrow [0, \infty]$ with $\mu(\emptyset) = 0$